

Project Title: Jamieh

Team Name: G3

Target Industry: EdTech

Team Members:

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The Core Problem (The "Why")

Static textbooks offer no structure, pacing, or accountability — learners are left to figure out what to study, when, and how deeply on their own. Meanwhile, those trying to switch careers or validate their existing skills have no way to test what they actually know versus what their CV claims. Who suffer the most are: Individual self-learners, career-switchers, and lifelong learners who lack access to structured academic environments or personalized skill validation.

1. The Solution & Value Proposition (The "What")

JAMIEH is an AI-driven learning simulator that converts any uploaded textbook into a comprehensive and fully featured university-style semester-long learning experience with weekly lectures, schedules, and attendance. It also includes support for multiple presentation formats and audio-visual presentation. In addition, users can upload their CV and select a target job role to receive an AI-driven skill gap analysis and recommendations for learning and validation quizzes to verify their skills.

- JAMIEH converts any textbook into an engaging and immersive university-style learning experience, replacing passive reading with an active and engaging experience. AI agents automatically create a syllabus, provide lectures from the textbook with proper citations through RAG, and enforce discipline through a schedule and penalties. It also provides real-time tutoring. In addition, for CV holders, an AI agent provides an analysis of claimed skills against a target role and recommends learning through JAMIEH and quizzes to verify actual skills.

- Convert any textbook into a comprehensive and fully featured university-style self-paced learning experience with accountability, multiple presentation formats,

and audio-visual presentation, and AI-driven skill validation without the need for a human instructor.

2. The AI Trinity Strategy (The "How")

The three pillars form a moat by combining LLM reasoning for intelligent curriculum design and content generation, RAG grounding to ensure every lecture and quiz is citation-accurate from the user's own textbook, and autonomous agents that orchestrate the entire academic lifecycle end-to-end — from syllabus planning to exam grading to CV-based skill validation — without any manual intervention. No existing platform delivers all three together in a self-contained learning simulator.

- **LLM (Intelligence):** GPT-5.4 / Claude

Reasoning, curriculum structuring, lecture generation, summarization, quiz creation, CV skill extraction, and gap analysis

- **RAG (Knowledge):** User-uploaded textbooks (PDF) chunked and embedded into pgvector for citation-grounded lecture retrieval, and parsed CV documents used for skill extraction and gap analysis against target job roles

- **Agents (Action):**

- Dean Agent (Curriculum Planner)

Automatically analyzes the uploaded textbook's table of contents and generates a full semester schedule—assigning weekly lectures, labs, and exams across the entire curriculum without user intervention.

- Lecturer Agent

Retrieves relevant textbook passages via RAG hybrid search (keyword + vector) and autonomously produces citation-grounded lecture scripts with page references for each scheduled session.

- Proctor Agent

Automatically generates quizzes and exams aligned with the weekly curriculum, ensuring questions are grounded in the textbook content covered that week.

- Grader Agent

Automatically evaluates student answers against textbook-grounded rubrics, provides detailed feedback, and updates the learner's progress tracking.

- TA Agent (Teaching Assistant)

Provides real-time tutoring by answering student questions during lectures using RAG-retrieved textbook content with citations.

- CV Analyzer Agent

Parses an uploaded CV, extracts claimed skills, validates them against a user-selected target job title, performs gap analysis, and suggests matching pre-made learning paths within JAMIEH to close identified skill gaps.

- CV Assessor Agent

Generates targeted quizzes and skill-validation exams based on the CV-claimed skills to verify the learner's actual proficiency level before assigning a learning path.

- Attendance Enforcement Agent

Automatically tracks student participation across scheduled lectures and labs, enforcing attendance policies and managing the logic for absences or required interventions.

3. Market & Competition

- **Primary User:** Individual self-learners, career-switchers, and lifelong learners who want structured, accountable education from any textbook and AI-validated skill verification for career advancement
- **The Advantage:** Unlike a search engine or generic course platforms (Coursera, Udemy), JAMIEH creates a full academic simulation grounded in the user's own textbook — with citation-backed lectures, enforced attendance discipline, multimodal delivery (audio + diagrams), automated grading, real-time tutoring, and CV-based skill validation. No existing tool combines structured curriculum generation, RAG-grounded content, and autonomous multi-agent orchestration in a single learning experience.

4. Execution Plan (High Level)

- **Core Stack:** (Nextjs / FastAPI or Django / PostgreSQL + pgvector)
- **Risk Factor:** Achieving faithful retrieval of RAG responses (>85% citation grounding) across varying textbook formats, as well as controlling hallucination in LLM responses when generating lecture questions and assessments—since if an LLM generates information not present in the textbook, all academic credibility of such a system falls apart.